Chipathon 2025
AC3E-Chile team

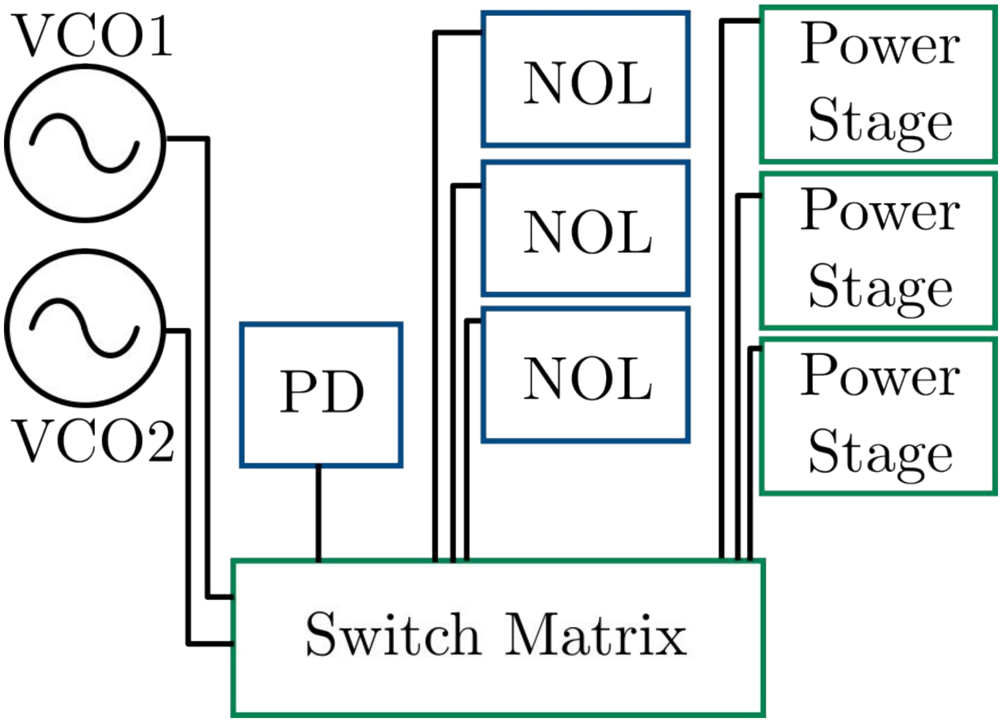
Power MOSBius - Project proposal

Datasheet - Functionality

- Power integrated circuits are a growing topic in the context of energy autonomous and compact electronic systems: IoT, wearables, nanosatellites, etc.
- At AC3E-USM, we have been researching and implementing power management integrated circuits using open source tools for the last 3 years
  - Easy to implement + modular + friendly with older technologies
- The project goal is to implement an educational chip for integrated power electronics - Power MOSBius -, focused on a full closed loop DC-DC buck converter operation test with digital and time-based control, including independent testing of each building block: open loop buck operation, VCO characterization and digital circuit testing

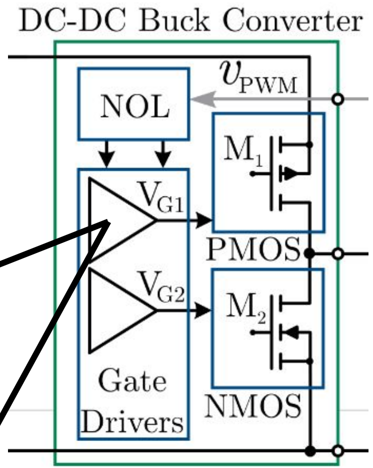
Datasheet - Block diagram

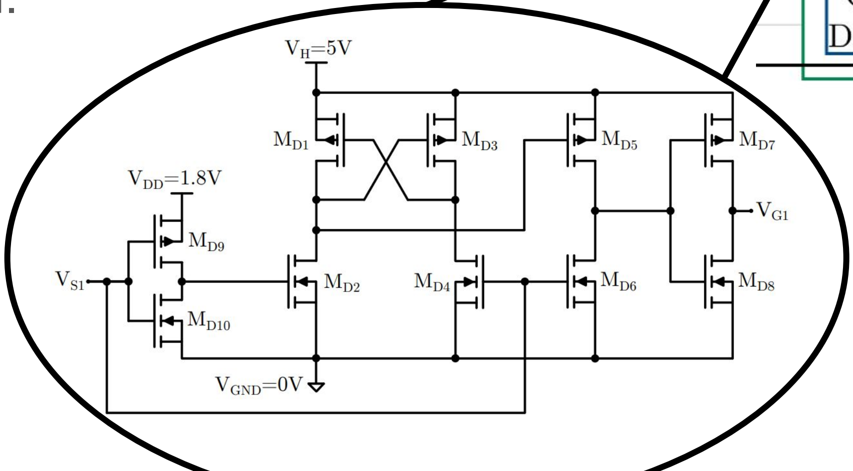
- To achieve this goal, the plan is to have a set of building blocks that, through the usage of the MOSbius switch matrix, will allow the implementation and testing of the different features



Datasheet - Block diagram (Power Stage)

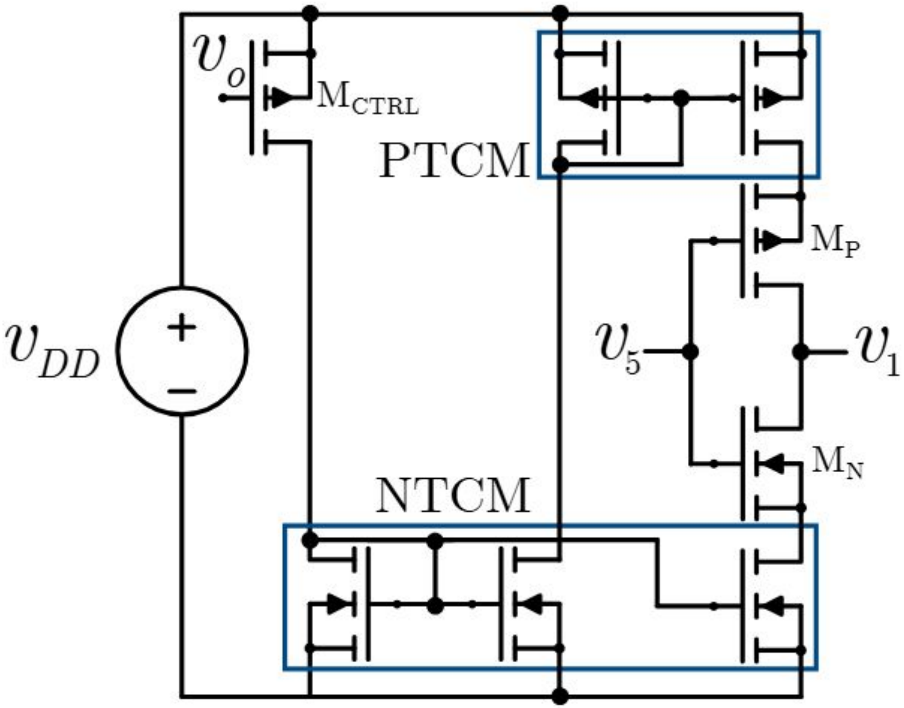
- Regarding complexity, the power stage has 2 transistors for a half bridge DC-DC converter and 10 transistors for each level shifter. Thus, 66 transistors for the 3 power stages involved in the chip. The design for each stage is quite complex due to the efficiency analysis and maximum current expected.





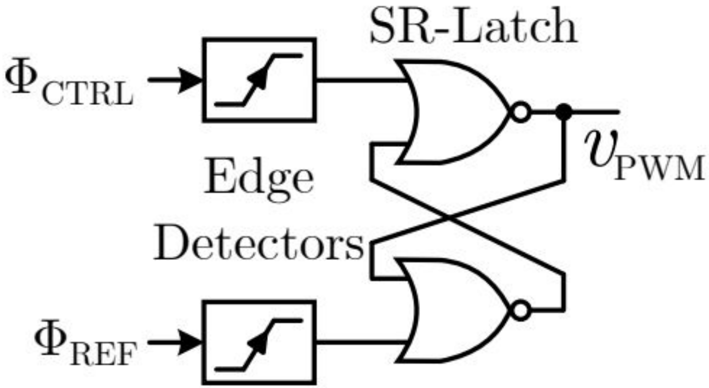
Datasheet - Block diagram (VCO)

- For the ring VCO, each stage will have 8 transistors. Depending on the phases required for the converters, the number stages for the ring will change using the switch matrix. The complexity for this design relays on the central frequency requirement and linearity, which is increases due to the amount of transistor parameters that affect it.

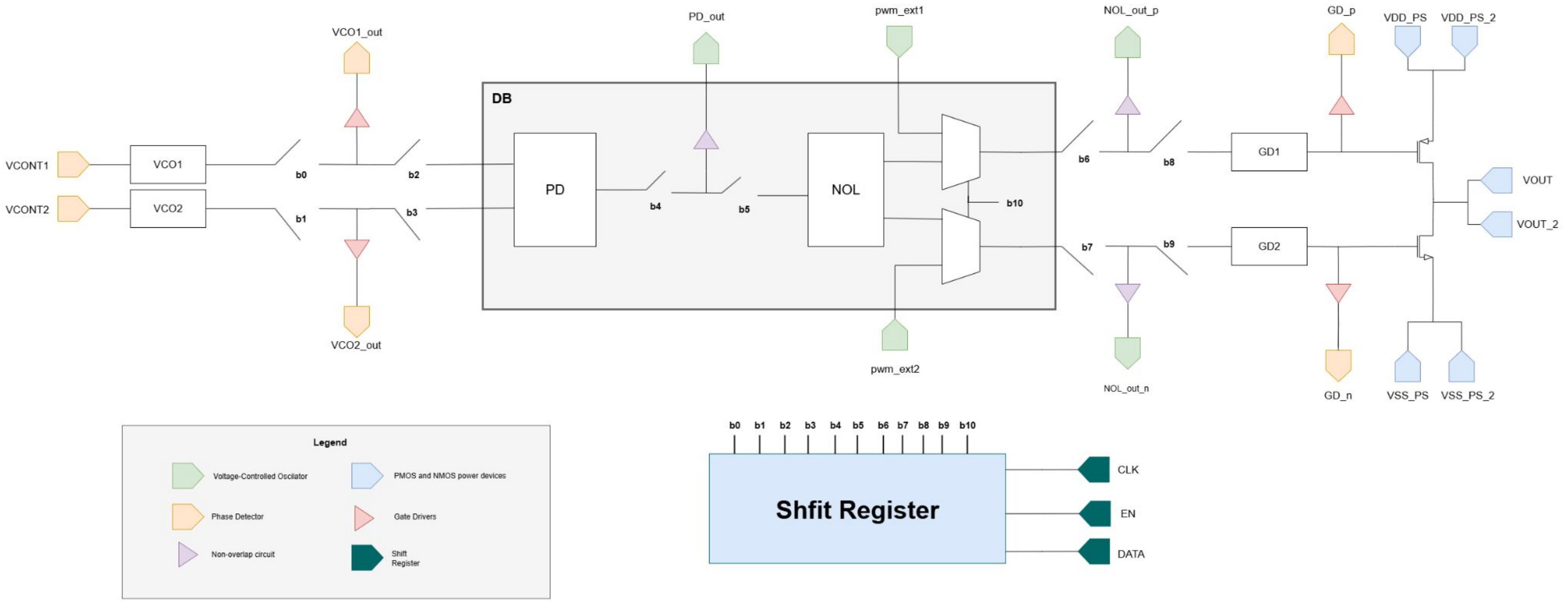


Datasheet - Block diagram (PD, NOL)

- For the Phase Detector (PD) and Non-Overlapping (NOL) blocks, the complexity is low due to the usage of standard cells from the GF180 library. Regarding the number of transistors, it is unknown at the moment as the standard cells haven't been explored internally.



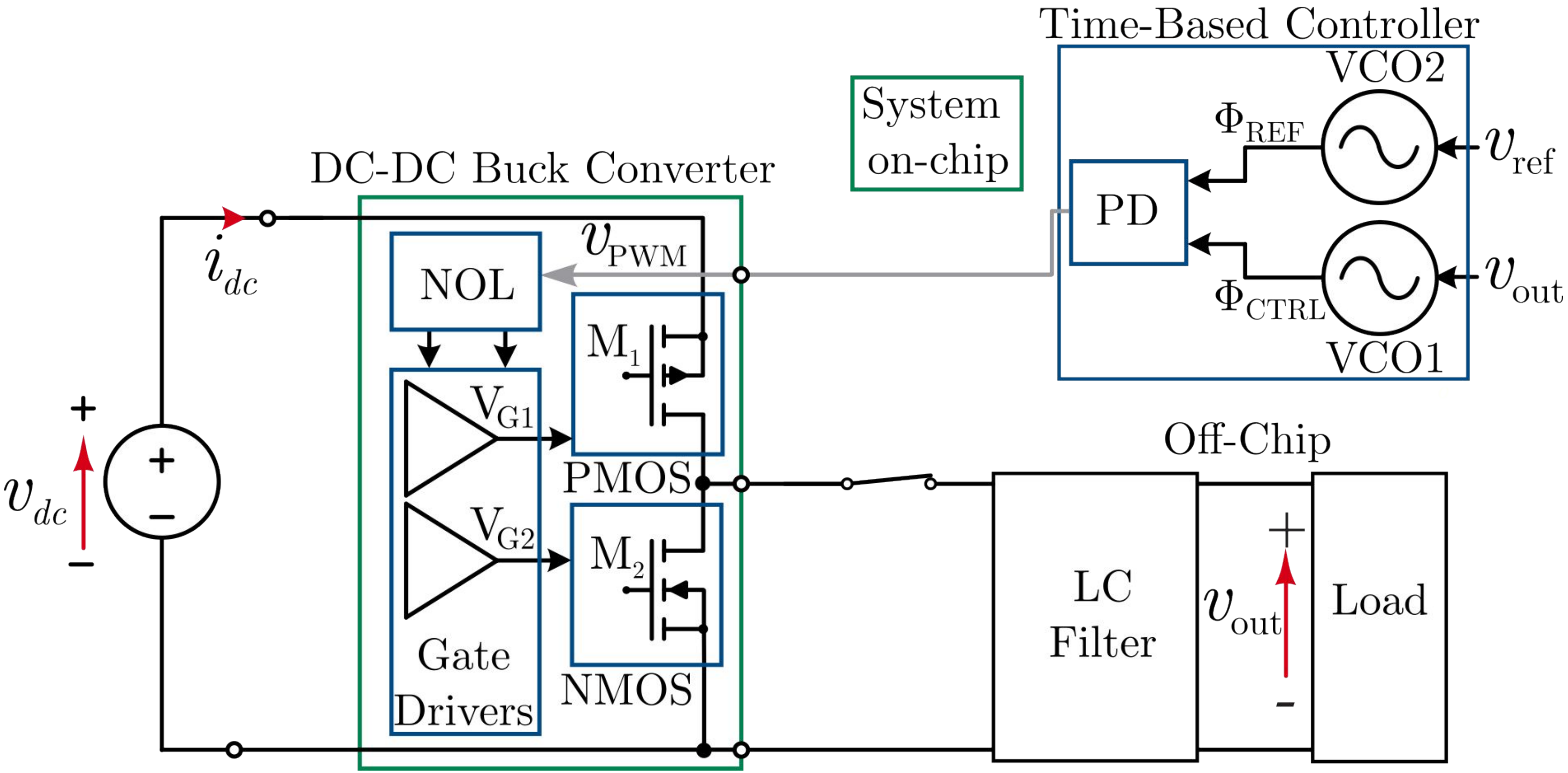
Datasheet - Pinout



Datasheet - Pinlist

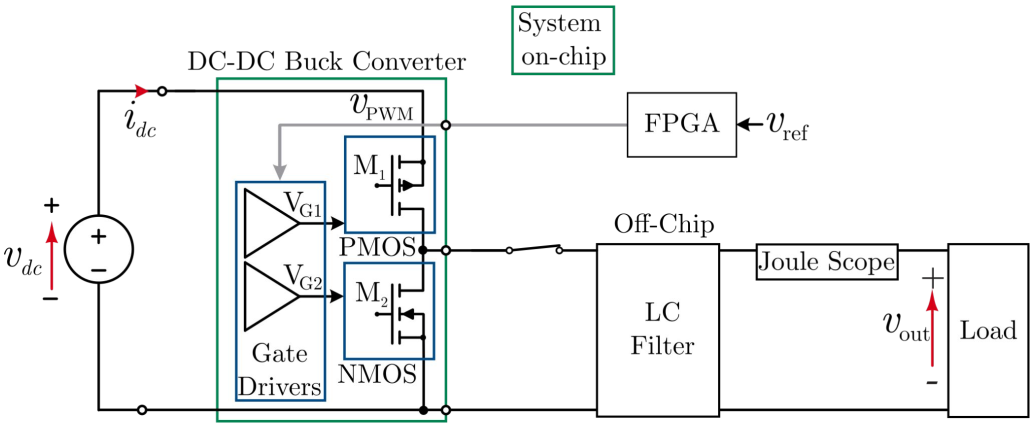
Pin #	Pin name	Direction	Type	Description
1	VCONT1	Input	Analog	Voltage-Controlled Oscillator control voltage
2	VCONT2	Input	Analog	Voltage-Controlled Oscillator control voltage
3	VCO1_out	Output	Analog	Output of Voltage-Controlled Oscillator
4	VCO2_out	Output	Analog	Output of Voltage-Controlled Oscillator
5	PD_out	Output	Digital	Output of Phase Detector
6	NOL_out_p	Output	Digital	Non-Overlapping Circuit control signal- positive side
7	NOL_out_n	Output	Digital	Non-Overlapping Circuit control signal- negative side
8	pwm_ext1	Input	Digital	PWM external input
9	pwm_ext2	Input	Digital	PWM external input
10	GD_p	Output	Analog	Gate driver control signal for PMOS (positive side)
11	GD_n	Output	Analog	Gate driver control signal for PMOS (negative side)
12	VDD_CTRL	Bidirectional	5V	Power Supply Voltage for PD,NOL and VCO
13	VSS_CTRL	Bidirectional	GND	Ground reference for PD,NOL and VCO
14	VDD_PS	Bidirectional	5V	Power Supply Voltage for the power stage
15	VDD_PS_2	Bidirectional	5V	Power Supply Voltage for the power stage
16	VSS_PS	Bidirectional	GND	Ground reference for the power stage
17	VSS_PS_2	Bidirectional	GND	Ground reference for the power stage
18	VOUT	Output	Analog	Output Voltage
19	VOUT_2	Output	Analog	Output Voltage
20	CLK	Input	Digital	Shift register Clock inputl
21	EN	Input	Digital	Shift register manual enable
22	DT	Input	Digital	Shift register Data input

Datasheet - Application diagram: single-phase operation

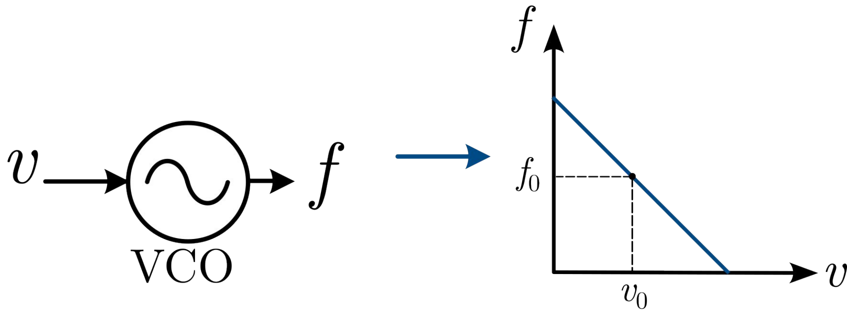


Datasheet - Application diagram: single-block testing

1. Open-loop operation



2. VCO characterisation



3. Testing of digital blocks

NOTES ON FPGA:

- Model: Nexys A7 FPGA (used in previous work)
- Why FPGA? Need short dead time values for open-loop op.
- FPGA will also be used for switch matrix control to avoid several boards

Team members + tasks

- Power stage + gate driver design
  - Maximiliano Jofré + Felipe Rojas
- Switch matrix layout macros
  - Fernanda Quintana + Max Vega
- Digital circuit (phase detector + dead time) design and automated layout
  - Jesus Ávila
- VCO design, top level simulations and proposal writing
  - Vicente Osorio + Jorge Marín
- Top integration
  - All members

Progress tracker

https://docs.google.com/spreadsheets/d/1zdIPwZ65STnkTxjjyIoI5YgF0EQ0I_z8IdRqjS
yOORw/edit?usp=drive_link

Week-by-week schedule

	34	* PS + GD layout	* Integration of Tgate array + digital circuits	* VCO top layout		* Help in top integration
	35	* PS + GD layout	* Integration of Tgate array + digital circuits	* Help in top integration	* Help in top integration	* Help in top integration
	36	* Help in top integration	* Tgate + digital co-simulation	* Help in top integration	* Help in top integration	* Help in top integration
	37	* Help in top integration	* Tgate + digital co-simulation	* Help in top integration	* Help in top integration	* Help in top integration
	38	* Help in top integration	* Help in top integration	* Help in top integration	* Help in top integration	* Help in top integration
	39	* Help in top integration	* Help in top integration	* Help in top integration	* Help in top integration	* Help in top integration
Layout and Verification	**40**	* Help in top integration	* Help in top integration	* Help in top integration	* Help in top integration	* Help in top integration